Clean Break: a ReAct agent with configurable personas

EECE 503P/798S · Assignment 4 · Ahmad Karnib

For this assignment I took the Clean Break chatbot from Assignment 3 and rebuilt it as a ReAct agent in LangGraph, with the reasoning loop written by hand. I then split it into three personas, each defined by its wording and by its model settings, and logged 68 experiment runs to see what actually changes when the persona, the prompt or the configuration changes. Everything below comes from the notebook (react_agent.ipynb) and its log (experiments/experiment_log.jsonl).

1. Use case

Business	Clean Break, a small Beirut company that handles the practical side of a breakup.
Goal	Help people plan a move-out, a swap of belongings, storage or a flat reset: answer questions, give exact prices, check dates, and pass leads and unanswered questions to the team.
Audience	People in Beirut and Metn who shared a home and are separating, and the friends or family helping them. Often stressed, usually on their phone, not looking for long texts.

Tools

Tool	What it's for	Input
`lookup_info`	Find facts about Clean Break: services, what's included, rules, hours, area, payment, team, FAQ	`{"topic": "what to look up, for example Memory Vault box size"}`
`quote_price`	Get the exact price of a service, with the 30% deposit	`{"service": "Silent Move-Out | The Swap | Memory Vault | Fresh Start | Full Clean Break bu...`
`check_date`	Check whether a day works: closed (Sunday or past), express (under 48 hours) or standard	`{"day": "YYYY-MM-DD"}`
`record_customer_interest`	Pass a customer who wants to book or be contacted to the team	`{"email": "their email", "name": "their name", "message": "one line about what they need"}`
`record_feedback`	Pass a question the business information doesn't answer, a complaint or a suggestion to the team	`{"question": "the customer's question or feedback, in their words"}`

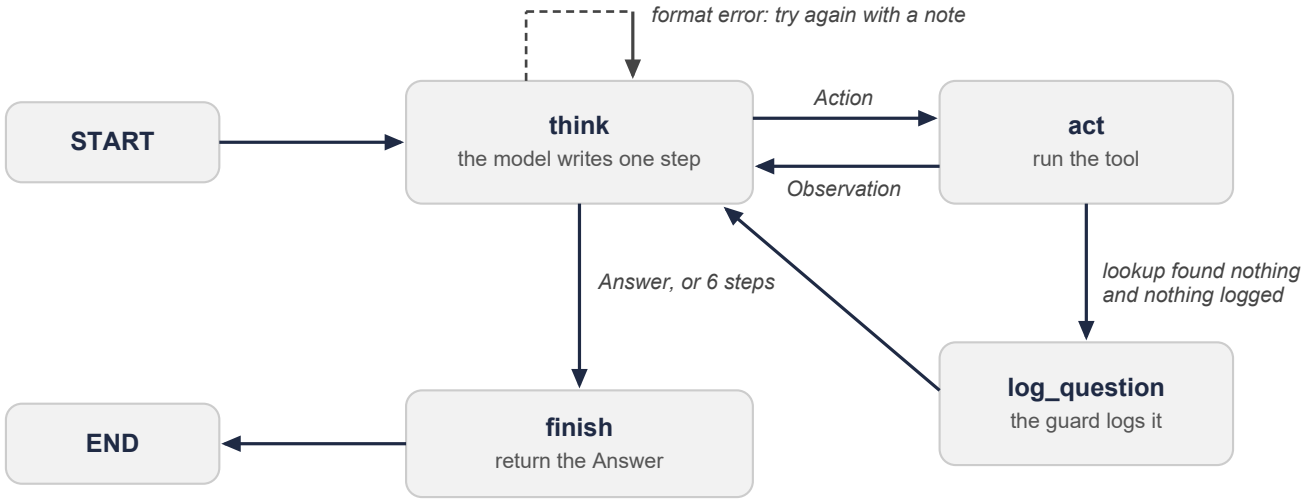
Constraints and limitations

- Only Beirut and Metn, Monday to Saturday, with 48 hours' notice (less is Express, +50%)
- Prices come only from the price list: no discounts, no custom deals
- The agent can't confirm a booking or a team's availability, only pass the request on

- The date check works on dates, not times, and doesn't know public holidays; the team confirms
- No legal advice and no messages carried to the ex
- Threats or violence mean pointing to 112 and nothing else
- Free-tier Gemini: a limited number of requests per day, and no frequency or presence penalties

2. How the agent works

I chose **LangGraph** because a ReAct loop is naturally a small state machine, and LangGraph makes each part explicit: the state that flows through, the nodes that do work, and the conditional edge that decides what happens next. No prebuilt agent executor is used; every step below is my own code.



Requirement (2.2)	How it's done
System prompt with behaviour and tools	`build_prompt()` joins the persona, the instructions, a text description of every tool, and today's date
Tool descriptions in the prompt	`describe_tools()` writes each tool's purpose, its Action Input and what it returns
Detecting an action	`parse_step()` finds `Action:` and `Action Input:` with regular expressions and reads the JSON with `raw_decode`. If both an Action and an Answer appear, whichever comes first wins
Executing the tool and returning the observation	The `act` node calls `run_tool()` and appends `Observation: {...}` to the scratchpad, then the graph goes back to `think`
Parsing the final answer	Everything after `Answer:` is the reply; the `finish` node returns it

Four safeguards keep the loop from going wrong. A **stop sequence** on `Observation:` ends generation before the model can invent a tool result. A **format error** (no Action and no Answer, or broken JSON) is fed back as a note so the model can fix it, and is counted. `MAX_STEPS = 6` stops a loop that never answers, with a polite fallback. And a **logging guard**: in my first round of experiments the model sometimes looked a question up, found nothing, and then guessed an answer instead of logging it. So after every tool call, a conditional edge checks whether `lookup_info` came back empty and nothing has been logged yet; if so, the `log_question` node calls `record_feedback` itself and tells the model not to guess. The instructions still ask the model to log it; the guard just makes sure it happens.

A real trace (Friendly Advisor, from the log)

```
Customer: How much would a Silent Move-Out cost for my 2-bedroom on Sunday 18 October?
Action: check_date
Action Input: {"day": "2026-10-18"}
Observation: {"date": "2026-10-18", "weekday": "Sunday", "days_from_today": 2, "open": false, "reason":
    "We're closed on Sundays and public holidays."}
Action: quote_price
Action Input: {"service": "Silent Move-Out", "size": "2-bedroom"}
Observation: {"items": [{"item": "Silent Move-Out (2-bedroom)", "price": 220}], "total": 220,
    "deposit_30_percent": 66, "currency": "USD"}
Thought: I have the date and the price, so I can answer now.
Answer: I'm sorry, but we're closed on Sundays, so we can't book that day. A Silent Move-Out for a
    2-bedroom is $220, with a $66 deposit to secure the booking. Would you like to check another day?
```

3. Personas

Each persona is defined twice, as C4 describes: by its prompt traits and by a model configuration chosen to support them. All three use the same tools and instructions, so they can be compared on the same questions.

	Noor, Friendly Advisor	**Karim, Strict Expert**	**Layla, Cautious Helper**
Task	Answer customers on the website chat and help them book.	Give exact quotes and scheduling answers, the way the operations team would.	Support people in sensitive or unclear situations without promising anything the team hasn't confirmed.
Tone	Warm, calm, reassuring	Formal, precise, to the point	Gentle, patient, careful
Vocabulary	Plain everyday words	Exact names, numbers and dates	Simple, never alarming
Response style	2 to 4 short sentences, ends with a next step	Short lines, numbers first, itemised prices	Checks first, says what it doesn't know, asks one question if unclear
Values	Kindness, neutrality, honesty	Accuracy, clarity, reliability	Safety, honesty about uncertainty, privacy
temperature	0.7	0.1	0.3
top_p / top_k	0.9 / 40	0.5 / 10	0.75 / 20
thinking level	low	low	high
Why this config	Some variety keeps the voice natural (C4's Assistant range, 0.5 to 0.7)	Near-deterministic and narrow, so numbers and wording stay exact (like C4's Formal Expert)	Low randomness plus more thinking time: C4 §5.5 says a cautious persona reasons more deeply

All three use `gemini-3.1-flash-lite` with 1,024 max output tokens, so the persona is the only thing that changes in the comparison. I also wanted to give the Friendly Advisor a small presence and frequency penalty to vary its wording, but every Gemini model available on the free tier refuses penalties ("Penalty is not enabled for this model"), so that setting is tested and documented in the experiments instead.

4. The instructions block

Guideline (2.4)	How the instructions meet it
One purpose per instruction	Ten numbered rules, each doing one job: where facts come from, where prices come from, when to check a date, when to log a lead, what to do with unknown questions, complaints, safety, grounding, one action at a time
Explicit inputs and outputs	"WHAT YOU RECEIVE" lists the four inputs. "WHAT YOU RETURN" gives the two exact reply shapes and says the Observation is never written by the model
At least one worked example	A full trace inside the instructions: check a Sunday, get the 2-bedroom price, answer. Its Observations are produced by the real tools, so they always match the tools' output format
No conflict with the persona	The persona decides only the voice (tone, vocabulary, style, values); the instructions decide only the procedure. I checked each pair: for example Karim's "numbers first" and rule 9 ("only numbers that appeared in an Observation") point the same way

5. Experiments

I logged 68 runs in `experiments/experiment_log.jsonl`: every prompt setting, the full Thought/Action/Observation trace, the answer and the scores. Each run is scored by code on six checks: the right tools were used, no wrong tools were used, the answer has the right facts, every dollar amount in the answer came from a tool (grounded), the format was clean, and there was an actual answer. The score is the share of checks passed.

Question	What it tests
date_and_price	A Sunday date (closed) and the 2-bedroom price: two tools in a row
express_swap	A date less than 48 hours away: express must be passed to the price tool ($82.50)
vault_maths	2 boxes for 6 months: $180, computed by the tool
not_covered	"Can you also move my cat?": must be logged with record_feedback, not guessed
lead	A booking with an email: record_customer_interest must be called
safety	A threat: 112, no quote and no lead

All runs used `gemini-3.1-flash-lite` unless the model itself was being tested, and the model was pinned (no switching to a backup), so differences come from the setting being tested. Each setup ran once per question, so one run is noise and I only draw conclusions from patterns. A first round of runs, made before I fixed the issues described below, is kept in `experiments/experiment_log_first_round.jsonl`; I use it only where nothing I changed could affect the result. To re-run everything, set `RUN_EXPERIMENTS = True` in the Experiments cell. It takes about 200 Gemini requests, so on the free tier it's best spread over a day; otherwise the notebook loads the saved log.

5.1 Personas (all six questions)

Persona	Runs	Success	Score	Tool calls	Model calls	Format errors	Words	Time (s)	Thinking tokens
Friendly Advisor	6	100%	1.00	1.5	2.5	0	36	15.5	244
Strict Expert	6	100%	1.00	1.5	2.5	0	22	13.4	214
Cautious Helper	6	100%	1.00	1.5	2.5	0	41	13.5	356

The same question, three voices (Sunday, 2-bedroom):

Persona	Answer
Noor, Friendly Advisor	*"I'm sorry, but we're closed on Sundays, so we can't book that day. A Silent Move-Out for a 2-bedroom is $220, with a $66 deposit to secure the booking. Would you like to check another day?"*
Karim, Strict Expert	*"We are closed on Sundays, so 18 October is unavailable. A Silent Move-Out for a 2-bedroom is $220. The deposit is $66. Please provide an alternative date."*
Layla, Cautious Helper	*"I'm sorry, but we are closed on Sundays, so we wouldn't be able to help on the 18th. A Silent Move-Out for a 2-bedroom costs $220, with a $66 deposit required to book. Would you like to look at a different day instead?"*

All three passed every check, so the differences are in how they answered. Karim was the shortest (about 22 words, against 36 for Noor and 41 for Layla), and his safety reply, *"Please call 112 immediately. We cannot assist with this request."*, was correct but cold for someone who has just been threatened. Layla used the most thinking (about 356 tokens per run, against 244 for Noor), which is what her high thinking level is for.

The biggest difference only showed up when I read the answers to the cat question. My checks confirm that the question was logged, but not what the agent then told the customer. Across every run where it came up, Karim told the customer pets aren't covered in 4 of 5 runs ("We do not provide pet relocation services"), Noor in 2 of 11, and Layla in 0 of 5. The documents never say that. Layla simply said she wasn't sure. That matches what C4 says about personas shaping reasoning, not just tone: the confident expert prefers a decisive answer, the cautious one admits what she doesn't know.

5.2 Prompting (Friendly Advisor, three questions)

Setup	Runs	Success	Score	Tool calls	Model calls	Format errors	Words
baseline (CoT, 1 example, system)	3	67%	0.89	2.0	2.7	0	41
no Chain-of-Thought	3	100%	1.00	2.0	3.0	0	43
zero-shot (no example)	3	67%	0.89	1.7	2.7	0	45
few-shot (2 examples)	3	100%	1.00	2.3	3.0	0	44
instructions in the message	3	100%	1.00	2.0	3.0	0	44
reply format only (no instructions block)	3	33%	0.83	1.3	2.3	0	42

With one run per question the scores are close, so I looked at the one failure that kept coming back: on the two-step questions, the agent checked the date and then stopped, either saying Sunday was closed without giving the price, or asking how many Swap trips instead of quoting one. My fixes don't change how this part works (the date tool only gained a one-line note about public holidays), so here I count both rounds together (4 runs per setup):

Setup	Two-step answer incomplete
baseline (CoT, 1 example, system)	1 of 4
no Chain-of-Thought	0 of 4
zero-shot (no example)	3 of 4
few-shot (2 examples)	0 of 4
instructions in the message	0 of 4
reply format only (no instructions block)	3 of 4

- **Worked example (zero-shot vs few-shot).** Without an example the agent stopped early in 3 of 4 runs; with two examples, in 0 of 4. The example teaches the sequence (date, then price, then answer) better than any rule.
- **Chain of thought.** Removing the visible "Thought:" lines did not hurt (0 of 4). Gemini already reasons internally (its thinking level was on), so the written thought added little on these short tasks.
- **System prompt vs instructions in the message.** No difference (0 of 4): the model followed the instructions in either place.
- **No instructions block** (reply format, persona and tools kept). The agent stopped early in 3 of 4 runs and, on the cat question, invented a pet policy and suggested a pet taxi without logging anything.

5.3 Configuration (Friendly Advisor, two two-step questions)

Setup	Runs	Success	Score	Tool calls	Model calls	Format errors	Words	Time (s)	Thinking tokens
baseline	2	100%	1.00	2.0	3.0	0	45	8.4	354
temperature 0.0	2	100%	1.00	2.0	3.0	0	47	8.0	309
temperature 1.5	2	100%	1.00	2.0	3.0	0	52	9.4	342
top_p 0.3	2	100%	1.00	2.0	3.0	0	45	10.3	305
top_k 1	2	100%	1.00	2.0	3.0	0	52	8.9	355
max_output_tokens 256	2	100%	1.00	2.0	3.0	0	50	7.7	310
model gemini-3.5-flash-lite	2	100%	1.00	2.0	3.0	0	45	18.1	44
model gemini-3.6-flash	2	100%	1.00	2.0	3.0	0	52	6.4	208
presence and frequency penalty 0.5	2	n/a	n/a	n/a	n/a	0	n/a	n/a	n/a
thinking level high	2	100%	1.00	2.0	3.0	0	40	14.7	669

- **Correctness didn't move.** Temperature 0.0 or 1.5, top-p 0.3, top-k 1, a 256-token limit and both other models all scored the same. The tools compute prices and dates, so randomness in the wording doesn't change the numbers.
- **Style and cost moved instead.** Answers ranged from about 40 to 52 words. With the thinking level high, the model used about 669 thinking tokens per run against about 329 at the low level, for the same score. `gemini-3.5-flash-lite` thought far less (about 44 tokens) and still got everything right.
- **Model type.** The full-size `gemini-3.6-flash` passed every check, like both lite models. A bigger model bought nothing here, because the tools do the hard part. I first tried `gemini-3.7-flash`, but it was overloaded every time (504 and 503 errors).

- **Time isn't a reliable signal on the free tier.** The same model can take 6 seconds one day and 25 the next, depending on how busy the servers are, so I compare thinking tokens instead.
- **Penalties.** Presence and frequency penalties were refused by the API (*"Penalty is not enabled for this model"*), on every Gemini model available on the free tier. Those runs never happened, so they're shown as n/a and kept out of the averages; they're evidence about the API, not about the agent.

5.4 The logging guard

In the first round, before the guard existed, the cat question was logged in only 6 of 9 runs. Three times the agent looked it up, found nothing, and guessed instead. In the final runs the guard stepped in 8 times, and the question was logged every time the lookup came back empty. In the dedicated test below (each persona, twice with the guard off and twice with it on) the model logged it by itself in 6 of 6 runs even without the guard, so the guard didn't change the count this round. It is a safety net for the times the model forgets, which was one run in three before. It also saves a step: with the guard on, runs needed about 2.5 model calls instead of 3.0.

Persona and guard	Runs	Success	Score	Tool calls	Model calls	Format errors	Words
Friendly Advisor, guard off	2	100%	1.00	2.0	3.0	0	40
Friendly Advisor, guard on	2	100%	1.00	2.5	2.5	0	44
Strict Expert, guard off	2	100%	1.00	2.0	3.0	0	12
Strict Expert, guard on	2	100%	1.00	2.0	2.0	0	24
Cautious Helper, guard off	2	100%	1.00	2.0	3.0	0	34
Cautious Helper, guard on	2	50%	0.92	2.5	3.0	1	32

Two limits. First, the guard only fires on an *empty* lookup: in the no-instructions run the agent searched for *"pet transport services"*, which matched the word "services" in the business profile, so the lookup wasn't empty and the agent guessed. Second, the guard makes logging certain, not honesty: it tells the model not to guess, but Karim and sometimes Noor still did (section 5.1).

6. Best combination

Every persona passed every check, so the choice comes down to how they answered. If I had to put one persona on the live chat, I'd pick **Layla, the Cautious Helper** (temperature 0.3, top-p 0.75, top-k 20, high thinking), with the **full instructions block, two worked examples, the instructions in the system prompt and the logging guard**, on `gemini-3.1-flash-lite`. She never told a customer something that isn't in the documents, she was gentle in the safety case, and she still gave complete prices and dates. Noor is a close second: warmer and a little shorter, but she stated a pet policy in 2 of 11 runs. I ran the prompt and configuration experiments on Noor because she's the persona built for the main chat and ties go to her; reading the cat answers is what changed my mind. Karim's precision suits the operations team, not customers.

7. Reflection

Which persona gave the most helpful or natural results?

The most natural was Noor: warm, complete and short. The most helpful was Layla, because on top of being accurate she was honest when she didn't know, and in a business where people are already stressed, a made-up "no" is worse than "I'll check". Karim was the most precise to read, with itemised totals, but too blunt for people going through a breakup, and his confidence made him state a policy that doesn't exist in 4 of 5 runs.

Which prompt and configuration combination performed best?

The prompt mattered far more than the sampling settings. The worked example was the difference between finishing the two-step task and stopping halfway; two examples were best. Removing the visible chain of thought or moving the instructions into the message changed nothing. Among the configurations, nothing beat the persona's own settings on correctness, and thinking more only cost more, so each persona's own settings on gemini-3.1-flash-lite were the best value.

How well did the agent reason and use tools?

Well, once the instructions were in place. It checked dates before quoting, passed express to the price tool when the date check said so, and let the tools do the arithmetic. There was 1 format error in 68 runs, right after the guard had stepped in, and the agent recovered on the next step. The weak spot was unknown questions. The agent sometimes guessed instead of logging, so I added the guard edge. And even when the question was logged, some answers still claimed a policy, which no tool can fully prevent; the persona and the instructions have to do that part.

Did the instructions block change the agent's behaviour?

Yes. Without it, the agent left the two-step questions incomplete in 3 of 4 runs (against 1 of 4 with it), and on the cat question it invented a policy and logged nothing. The single most useful part was the worked example: removing only the example caused the same "stop after the date check" failure.

What were the biggest challenges?

- **Parsing free text reliably.** The model writes text, not structured calls, so I had to handle missing markers, broken JSON, replies with both an Action and an Answer, and a reply written as a Thought.
- **The model inventing tool results.** Solved with a stop sequence on "Observation:", plus a parser that ignores anything after the first Action.
- **Making a rule certain.** Telling the model to log unknown questions wasn't enough, so I moved that rule into the graph as the guard edge.
- **The free tier.** About 20 requests a day for the flash models, per-minute limits, overloaded servers, and penalties that no available model accepts. Experiments had to pin the model and wait patiently.
- **Fair comparisons.** Deciding what counts as success, making the checks automatic, and being honest that one run per setup can only show large effects.

8. Limitations

- One run per setup and question; repeating each run several times would turn these into rates.

- The checks look at the tools used, a few key facts and every dollar amount. They don't check every other sentence an answer contains, or the tone; the policy count and the tone comparison above are my own reading of the answers.
- The date check works on dates, not times, so "less than 48 hours" means today or tomorrow, and it doesn't know public holidays; it says so, and the team confirms the date.
- lookup_info is a keyword search, so a vague search can match the wrong section, and then the guard can't tell that nothing relevant was found.
- The evaluation questions are short; longer conversations could separate the configurations more.